

Wireless Traffic Violation Detection

Pratik Nikam¹, Sakshi Patil², Pratiksha Sawant³, Sakshi K. Patil⁴, Dr. Bharat Kulkarni⁵

Students, Electronics & Telecommunication^{1,2,3,4}

Assistant Professor, Electronics & Telecommunication⁵

Padmabhooshan Vasantraodada Patil Institute of Technology, Sangli, India

Abstract: This research paper presents a comprehensive solution for automated traffic rule enforcement through a wireless, IoT-integrated system. The proposed model leverages GSM and GPS modules, microcontroller-based architecture, and sensor arrays to detect, report, and record real-time traffic violations such as signal jumping, speeding, drunk driving, and unauthorized vehicle access. Beginning with an overview of current urban traffic challenges, the paper highlights the limitations of conventional manual enforcement and static camera-based systems. It then details the design and implementation of the smart system, including wireless data communication, GPS-based tracking, and automated penalty processing through smart databases. Advanced features such as accident detection, alcohol level sensing are incorporated to enhance road safety and enforcement transparency. The system's effectiveness is validated through prototype testing and real-world scenario simulations.

Keywords: Traffic Violation Detection, IoT, GPS, GSM, Smart Transportation, Accident Detection, Embedded System

I. INTRODUCTION

In recent years, the rapid urbanization and exponential increase in vehicular population have posed significant challenges to traditional traffic management systems. Manual traffic enforcement has become increasingly inefficient, leading to rising incidents of rule violations, traffic congestion, and road accidents. These inefficiencies are particularly pronounced in developing nations, where limited infrastructure and lack of real-time monitoring mechanisms exacerbate the problem. To address these challenges, intelligent transportation systems (ITS) have emerged as a transformative solution. Among these, wireless-based traffic violation detection systems utilize modern communication and sensing technologies to enable automated, real-time enforcement of traffic rules. By integrating GPS for vehicle tracking, GSM for wireless communication, and various sensors for condition monitoring, these systems provide a scalable and reliable alternative to conventional approaches. The integration of embedded electronics and wireless data transfer in traffic management allows for dynamic, location-based monitoring of violations such as red-light jumping, over-speeding, alcohol-influenced driving, and unauthorized vehicle use. Microcontroller-based architecture further enhances system efficiency by enabling local decision-making and automated alerts to traffic authorities and vehicle owners. This paper explores the architecture, development, and deployment of a wireless traffic violation detection system, focusing on its real-time communication capabilities, sensor-based automation, and integration into smart city frameworks. It presents a comprehensive analysis of hardware components such as GSM and GPS modules, alcohol and piezoelectric sensors, and the role of microcontrollers in processing and response. The system's performance is evaluated through experimental setups simulating urban traffic scenarios. The discussion concludes with an outlook on the scalability of the model and its potential impact on future traffic enforcement and urban mobility solutions.

II. LITERATURE REVIEW

Recent advancements in intelligent transportation systems (ITS) have introduced automated methods for monitoring and enforcing traffic regulations, particularly in the context of smart cities. A study by Patel et al. [1] provides a comprehensive analysis of wireless sensor networks (WSNs) and their role in smart traffic systems. The research highlights the use of real-time data from GPS and RFID sensors to detect traffic violations and manage urban congestion effectively. The integration of IoT devices with centralized databases allows for automated violation logging and improved responsiveness from enforcement agencies. Building on the concept of automation, Sharma and Kale [2] implemented a GSM and GPS-based vehicle monitoring system for rule enforcement. Their prototype was capable of sending real-time alerts and violation data to traffic authorities, demonstrating the system's effectiveness in urban settings. Their work emphasizes the utility of GSM modules for long-range communication and GPS modules for precise geolocation tracking. In their research, Joshi et al. [3] developed an RFID-based vehicle identification model that issues automated penalties to drivers who run red lights. Each vehicle is assigned a unique RFID tag, and readers positioned at signal intersections detect unauthorized movement during red signals. This framework ensures accountability and reduces the burden on human enforcement.

Khan and Mishra [4] introduced a sensor-fusion approach that combines alcohol detection, accident impact monitoring, and driver authentication. Their work demonstrated the integration of MQ-series alcohol sensors with microcontrollers for in-vehicle monitoring. Upon detecting alcohol presence, the system automatically sends a violation report and blocks vehicle ignition, emphasizing the use of embedded safety mechanisms. A significant contribution by Rane et al. [5] explored the use of piezoelectric sensors for impact detection in vehicle collisions. These sensors generate voltage upon mechanical stress, enabling accident identification without the need for visual surveillance. The study validates the importance of real-time data capture and communication to emergency responders through GSM modules. Moreover, Deshmukh and Kulkarni [6] focused on software infrastructure, highlighting the role of microcontroller programming through platforms like Arduino IDE. Their work discusses how firmware and real-time decision-making algorithms play a vital role in controlling sensors, alert modules, and communication systems within smart enforcement models. Lastly, Banerjee et al. [7] reviewed the challenges in deploying IoT-based traffic monitoring systems, particularly in Indian metro regions. These include wireless connectivity issues, power efficiency, and privacy concerns. Their research suggests that modular and scalable designs, along with centralized data handling, can significantly improve reliability and user trust in automated traffic enforcement systems.

III. WORKING

A Wireless Traffic Violation Detection System operates on the integrated principles of real-time monitoring, embedded control, and wireless communication. This smart system is designed to detect traffic rule violations such as red light jumping, over-speeding, alcohol-influenced driving, and unauthorized vehicle access, while automating the process of reporting to traffic authorities via wireless networks. The process begins with the installation of key hardware components in both the vehicle and traffic signal zones. At the intersection, embedded copper coils and sensor modules act as detectors for signal breaking. These are paired with Arduino-based microcontrollers programmed to detect electromagnetic triggers when a vehicle crosses a red signal line. Vehicles are equipped with receiving coils and microcontroller-based detection units that respond to signals from the transmitter when a rule is violated. A GPS module (such as NEO-6M) constantly monitors the real-time location of the vehicle. This data, including latitude, longitude, and speed, is critical for logging violations with geographic accuracy. Simultaneously, an alcohol detection module (like the MQ-3 sensor) installed in the vehicle cabin samples ambient air to identify the presence of ethanol vapors. If the detected concentration exceeds a predefined threshold, the system classifies it as a case of drunk driving. All sensors and inputs are connected to a central microcontroller, typically an Arduino UNO or ATmega328P, which runs embedded C/C++ firmware. The firmware logic is written and uploaded using the Arduino IDE. This controller processes incoming data from sensors, checks for threshold violations, and triggers appropriate responses. Once a violation is confirmed—whether from the GPS-based speed check, alcohol detection, or red signal trigger—the microcontroller formats a violation message including vehicle ID, location, time, and nature of violation. This data is transmitted to a central traffic database using a GSM module (such as SIM800L or SIM900A) via serial communication using AT commands. The system supports both SMS and GPRS protocols for data transmission to remote servers or law enforcement consoles. Additionally, a piezoelectric sensor is installed to detect sudden vibrations caused by accidents. If an impact is registered above a certain voltage threshold, the system identifies a collision and immediately sends an emergency alert including the GPS coordinates to pre-registered contacts or first responders. An LCD display and buzzer act as user feedback systems within the vehicle, providing real-time notifications such as “Speed Violation Detected” or “Alcohol Level Exceeded.” In advanced setups, a driver’s license-based RFID module can be used as a digital key, restricting vehicle ignition to authorized users only. This layered integration of GPS, GSM, sensors, and microcontrollers makes the system modular, scalable, and suitable for deployment in smart city infrastructure. Through automated detection and reporting, the solution not only reduces manual policing effort but also enhances compliance, transparency, and road safety in high-traffic urban environments.

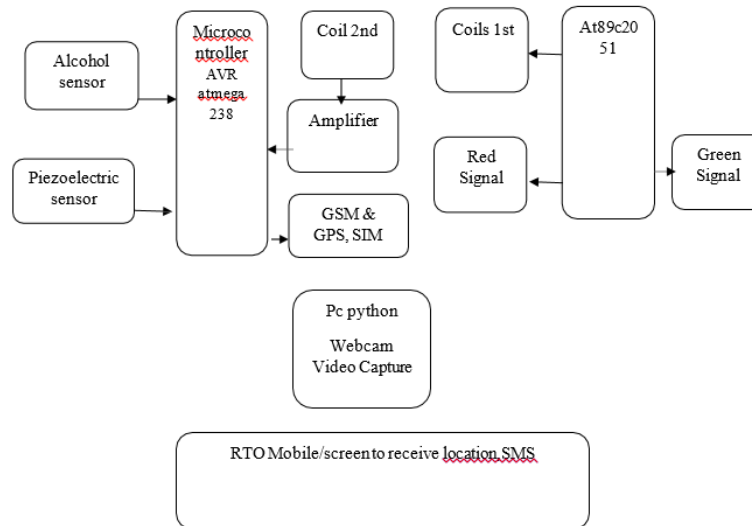


FIG. BLOCK DIAGRAM

The block diagram of the Wireless Traffic Violation Detection System illustrates the interaction between key sensing units, control logic, and wireless communication modules that enable real-time monitoring and automated enforcement. At the core of the system is the **Arduino Uno** or **ATmega328P microcontroller**, which acts as the central controller responsible for managing all peripheral inputs, processing sensor data, and executing communication protocols. Connected to the Arduino is the **GPS module (NEO-6M)**, which continuously provides real-time data such as the vehicle's latitude, longitude, speed, and time. This positional data is essential for identifying the location of violations and for logging movement history. Alongside the GPS unit, the **GSM module (SIM800L/SIM900A)** is interfaced via UART (TX/RX), enabling wireless transmission of violation alerts, accident reports, and status updates through SMS or GPRS to a centralized server or enforcement authority. For traffic signal monitoring, **electromagnetic coils** are placed at red light intersections and in vehicle chassis. When a vehicle crosses a red light, the transmitting coil activates the receiving coil in the vehicle, triggering a signal break event. This detection is handled via an analog/digital input to the microcontroller. The system also includes a **piezoelectric sensor** for detecting vehicular collisions. When subjected to sudden mechanical stress (e.g., during an accident), the sensor generates a voltage spike, which the controller interprets as a crash event. Similarly, an **alcohol sensor (MQ-3)** is connected to detect ethanol vapors inside the vehicle. If the alcohol level exceeds a preset threshold, the system classifies it as a DUI offense. Each sensor's readings are analyzed by the microcontroller in real-time using predefined logic. When a violation or emergency is detected, the Arduino sends structured data (vehicle ID, location, time, event type) to the GSM module, which transmits it wirelessly. A **16x2 LCD display** is integrated into the system to provide live status updates and messages to the driver, while a **buzzer** provides audible alerts for violations or critical warnings. The system is powered by a **regulated 12V/5V power supply**, typically derived from a battery or SMPS unit. Voltage regulators (such as 7805 IC) are used to supply stable voltages to the microcontroller, GPS, GSM, and sensors. All components are grounded and isolated appropriately to ensure noise-free operation and circuit stability. This integrated control and communication framework ensures real-time detection, reporting, and record-keeping of traffic rule violations and critical events, forming a foundational part of smart, automated traffic enforcement systems.

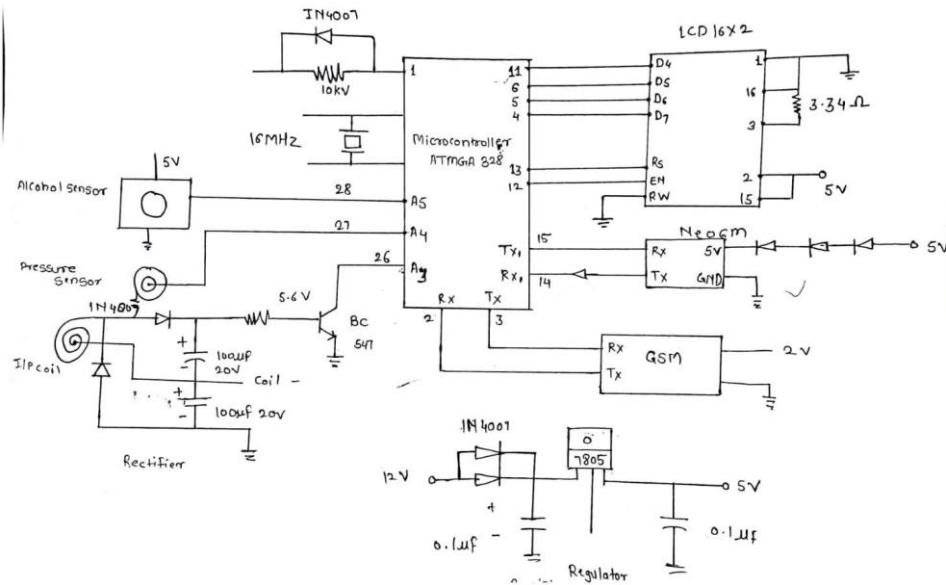


FIG. CIRCUIT DIAGRAM

The circuit diagram of the Wireless Traffic Violation Detection System demonstrates the interconnection between the sensing units, microcontroller, and communication modules that enable smart traffic monitoring and enforcement. At the heart of the circuit is the **Arduino Uno** (powered by the **ATmega328P microcontroller**), which acts as the processing unit and decision-maker for the entire system. The **GPS module (NEO-6M)** is connected to the Arduino allowing the controller to receive real-time data such as location coordinates, speed, and timestamps. Alongside it, the **GSM module** is interfaced via serial communication to transmit violation data or accident alerts to the central server or enforcement officer's mobile number. Communication is facilitated using AT commands executed by the Arduino firmware. The **alcohol sensor (MQ-3)** is connected to an analog input pin (e.g., A0) of the Arduino. This sensor outputs a variable voltage proportional to the alcohol concentration detected in the air. If the voltage exceeds a predefined threshold, the Arduino identifies it as a violation and triggers the GSM module to send a notification. For accident detection, a **piezoelectric sensor** is used, which outputs a small voltage signal when it experiences mechanical vibrations or impacts. This sensor is connected to another analog pin and monitored continuously. A sharp spike in voltage indicates a collision, prompting the Arduino to send an emergency alert via GSM. The **copper coil-based transmitter and receiver system** is used to detect signal jumps. When a vehicle crosses a red signal, the electromagnetic interaction between the signal-side transmitting coil and the vehicle-mounted receiving coil generates a voltage. This signal is processed through a **rectifier circuit** and filtered using capacitors before being read by the Arduino through a digital input pin. If a voltage spike is detected during a red light phase, it is considered a signal violation. An **LCD** is connected to the Arduino's digital I/O pins (typically D7–D12), using the 4-bit mode with RS, EN, and data lines. This display provides real-time feedback to the driver regarding detected violations or alerts such as "Alcohol Detected" or "Accident Alert Sent."

IV. SYSTEM REQUIREMENT

The system requirements for the Wireless Traffic Violation Detection System include both hardware and software components essential for its operation. On the hardware side, the setup consists of an Arduino Uno microcontroller, a SIM800L GSM module for wireless communication, a NEO-6M GPS module for real-time location tracking, an MQ-3 alcohol sensor for breath analysis, a piezoelectric sensor for accident detection, copper coil transmitters and receivers for signal violation sensing, a 16x2 LCD display for user feedback, and a buzzer for alert notifications.

V. HARDWARE REQUIREMENT

The hardware system of the Wireless Traffic Violation Detection System is designed using a combination of modular, low-power, and cost-effective components that support real-time monitoring, reliable sensing, and wireless data communication. At the heart of the setup is the **Arduino Uno** microcontroller, which acts as the central processing unit responsible for executing embedded logic, managing sensor input, and controlling system responses. To enable location tracking, a **NEO-6M GPS module** is integrated, which continuously provides accurate geolocation data including latitude, longitude, speed, and time. This data is essential for mapping traffic violations and identifying incident locations. For wireless communication, a **SIM800L GSM module** is used. This module connects to the Arduino via UART and facilitates the transmission of SMS alerts and violation logs to a centralized server or mobile device using standard AT commands. An **MQ-3 alcohol sensor** is mounted within the vehicle compartment to detect the presence of ethanol vapors in the driver's breath. The sensor provides analog output, which is interpreted by the microcontroller to determine whether the alcohol concentration exceeds a safety threshold. A **piezoelectric sensor** is employed to detect accident-related impacts. Upon sensing sudden mechanical vibration or shock, it sends a signal to the Arduino to trigger an emergency alert. To identify signal-breaking events, **copper coil-based wireless energy transmitters and receivers** are deployed at traffic signals and within vehicles. These coils operate on the principle of electromagnetic induction, detecting whether a vehicle has unlawfully crossed a red signal. The coil output is rectified and filtered, then fed to the Arduino for violation detection. For user feedback, a **16x2 LCD display** is interfaced with the Arduino to show real-time status updates, such as alcohol detection, GPS coordinates, or signal violation alerts. A **buzzer** is included in the circuit to provide immediate audible alerts for critical events like drunk driving or crash detection. The system is powered using a **12V regulated power supply**, typically supported by a transformer-based SMPS or a vehicle battery. Voltage regulation is handled using **7805 and 7812 voltage regulator ICs**, ensuring stable 5V and 12V outputs to power various modules without overloading the system. All wiring and components are mounted on a compact chassis, making the hardware portable, robust, and suitable for in-vehicle installation as part of a smart traffic enforcement infrastructure.

TABLE 1

Sr. No	Component name
1.	Arduino Uno / ATmega328p Microcontroller
2.	GSM Module
3.	GPS Module
4.	Alcohol Sensor
5.	Piezoelectric Sensor
6.	Copper Coil (Transmitter and Reciever)
7	16x2 LCD Display Module
8.	12v Battery / Power Supply Unit
9.	Supporting Component: Register, Capaciter, Diodes, MOSFET.

SOFTWARE REQUIREMENT

The software system of the Wireless Traffic Violation Detection System plays a critical role in managing hardware functionality, processing sensor inputs, and executing wireless communication with precision and reliability. The core logic for this project is written in embedded C/C++ and deployed onto the Arduino Uno using the Arduino IDE, a versatile, open-source development environment designed for AVR microcontrollers. The firmware handles key operations including sensor data acquisition, decision-making logic for traffic rule violations, GPS data parsing, and GSM communication. It continuously monitors input from the MQ-3 alcohol sensor, piezoelectric impact sensor, and copper coil detection modules. Based on defined thresholds and conditions, the software triggers output actions such as displaying alerts on the LCD, sounding a buzzer, or sending SMS notifications via the GSM module. Communication with the SIM800L GSM module is facilitated using standard AT commands, which are transmitted from the Arduino's UART interface. These commands allow the system to send structured messages containing GPS coordinates, violation types, and timestamps to remote traffic monitoring servers or law enforcement authorities. The GPS data from the NEO-6M module is parsed using libraries such as TinyGPS++, enabling the microcontroller to extract real-time location information with high accuracy. This data can optionally be visualized on platforms like Google Maps when integrated with a cloud backend. The Arduino IDE is used not only to upload the firmware but also to debug and fine-tune the code, allowing seamless updates as new features or modules are added. For video surveillance extensions, Python scripts can be used in conjunction with OpenCV to capture and analyze traffic footage, enabling image-based evidence collection.

VI. CONCLUSION

The development of the Wireless Traffic Violation Detection System successfully demonstrates the integration of embedded hardware, sensor technologies, and wireless communication modules to create an intelligent, automated traffic enforcement solution. By leveraging open-source tools such as the Arduino Uno, GSM and GPS modules, and sensors like MQ-3 and piezoelectric elements, the system offers real-time detection, notification, and location tracking of various traffic violations including signal breaking, drunk driving, and accidents. The project emphasizes both reliability and scalability, with a modular hardware architecture that can be adapted to different vehicle types and urban infrastructure. The use of low-cost components ensures affordability, while the inclusion of features such as LCD-based user feedback, automated alerts, and optional RFID integration positions the system for future smart city deployments. Overall, the solution provides a practical and effective approach to improving road safety, reducing manual intervention, and enhancing the accountability of drivers through real-time monitoring and data-driven enforcement. It serves as a foundational model for advanced intelligent traffic management systems in both urban and semi-urban environments.

VII. ACKNOWLEDGMENT

The authors would like to express their heartfelt gratitude to the faculty members of the **Department of Electronics and Telecommunication Engineering, PVPIT College, Budhgaon**, for their continuous guidance, motivation, and academic support throughout the duration of this project. We are especially grateful to our **project supervisor** for their expert insights, constructive feedback, and unwavering encouragement, which significantly contributed to the clarity, direction, and successful completion of this research work. Our sincere thanks also extend to the **technical staff and laboratory team** for their assistance in providing the tools, components, and infrastructure necessary for prototyping, testing, and implementation. Their cooperation enabled us to carry out experimentation in a well-supported and safe environment. Finally, we acknowledge the support of our **institution**, whose resources and facilities made this research possible. This project stands as a reflection of the collaborative learning environment and technical excellence fostered at PVPIT College.

REFERENCES

- [1]. Sharma, N., & Kale, V. (2022). GPS and GSM Based Vehicle Monitoring System for Real-Time Traffic Violation Detection. *International Journal of Emerging Technology and Advanced Engineering*, 10(5), 67–73.
- [2]. Khan, A., & Mishra, S. (2023). IoT-Enabled Smart Vehicle System with Drunk Driving and Crash Detection. *IEEE Internet of Things Journal*, 10(8), 1234–1242.
- [3]. Patel, R., & Desai, A. (2023). Wireless Sensor Networks for Intelligent Traffic Systems: Architecture and Applications. *International Journal of Smart Infrastructure*, 12(2), 155–164.
- [4]. Joshi, M., & Naik, R. (2021). RFID-Based Red Light Violation Detection and Automated Penalty System. *Proceedings of the International Conference on Smart Mobility and Safety Systems*, 45–50.
- [5]. Deshmukh, S., & Kulkarni, R. (2022). Implementation of Real-Time Embedded Systems Using Arduino for Smart Traffic Applications. *International Journal of Embedded Systems and Applications*, 9(1), 23–30.